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1.1 Recap

$$f(x+a) = e^{a\cdot\nabla} f(x)$$

1.2 Dipole

Start with 2 point charge

$$\phi(x) = \frac{Q}{4\pi\varepsilon_0|x-d|} + \frac{-Q}{4\pi\varepsilon_0|x|}$$

When the charges are close together, we can approximate the potential as a dipole potential

$$|d| \ll |x|$$

Writhe the potential as

$$\phi(x) = (e^{-d\cdot\nabla} - 1)\frac{Q}{4\pi\varepsilon_0|x|}$$

$$\begin{aligned}\phi(x) &= [(1 - d\cdot\nabla + \mathcal{O}(|d|^2))\frac{1}{|x|}] \\ &= \frac{Q}{4\pi\varepsilon_0}\left(-d\cdot\nabla\frac{1}{|x|} + \mathcal{O}(|d|^2)\right) \\ \implies Q(x) &= \frac{Q}{4\pi\varepsilon_0}\frac{d\cdot x}{|x|^3} + \frac{Q}{4\pi\varepsilon_0}\mathcal{O}(|d|^2)\end{aligned}$$

Take $Q\to\infty$, $d\to 0$ such that

$$p = Qd$$

is finite, we get the dipole potential

$$\phi(x) \to \frac{1}{4\pi\varepsilon_0}\frac{p\cdot x}{|x|^3}$$

1.3 Potential of a dipole

Zero charge but $E \neq 0$
Field close

$$E(x) = -\nabla\phi(x) = -\frac{1}{4\pi\varepsilon_0}\nabla\frac{3x(p\cdot x) - p(|x|^2)}{|x|^3}$$

A cube behaves like a dipole when the net charge is zero but there is a separation of positive and negative charges.

$$p = Qd$$

dipole moment
How does a diple respont to and applied electirc field E Use same limit as before, $Q\to\infty$, $d\to 0$ such that $p = Qd$ is finite, to find the response of the dipole to an applied electric field E. Total force on a dipole in an electric field is given by

$$\begin{aligned}F &= Q(E(x+d) - E(x)) \\ &= Qe^{d\cdot\nabla}E(x) - QE(x) \\ &= Q(d\cdot\nabla)E(x) + Q\mathcal{O}(|d|^2)\end{aligned}$$

Take the limit as before, we get

$$F = (p \cdot \nabla)E(x)$$

Work done moves a dipole

$$\begin{aligned} F &= -(p \cdot \nabla)\nabla\phi(x) \\ &= -\nabla(p \cdot \nabla\phi(x)) \end{aligned}$$

Work done on a dipole in an electric field is given by

$$\begin{aligned} W &= -\int_{x_2}^{x_1} dx \cdot F \\ &= -\int_{x_2}^{x_1} dx \cdot \nabla(p \cdot \nabla\phi(x)) \\ &= -(p \cdot \nabla\phi(x_1) - p \cdot \nabla\phi(x_2)) \\ &= -(p \cdot E(x_1) - p \cdot E(x_2)) \end{aligned}$$